

## Thermal physics · Mock midterm test · Version A · Solutions

Total 44 marks: I = 8, II = 14, III = 14, IV = 8. Marks for each part are shown on the right. A mark in square brackets [1] is one point earned by the step it follows. If you used a wrong value consistently in later steps (error carried forward), you still earn the later method marks. Section IV is answered in Italian.

### I · Multiple choice

[8]

| # | Answer                                                                     | Marks |
|---|----------------------------------------------------------------------------|-------|
| 1 | B — Heat is energy transferred because of a temperature difference.        | 1     |
| 2 | C — X and Y have the same temperature, even if they never touch.           | 1     |
| 3 | A — 300 K                                                                  | 1     |
| 4 | D — 0 K is the lowest temperature possible, so nothing can be colder.      | 1     |
| 5 | C — P is more sensitive: its liquid column moves further for each °C.      | 1     |
| 6 | B — 2.0 mm                                                                 | 1     |
| 7 | D — It increases, as if the hole were made of the same metal as the plate. | 1     |
| 8 | A — 0.30 %                                                                 | 1     |

## II · Paper 6 on the thermistor data

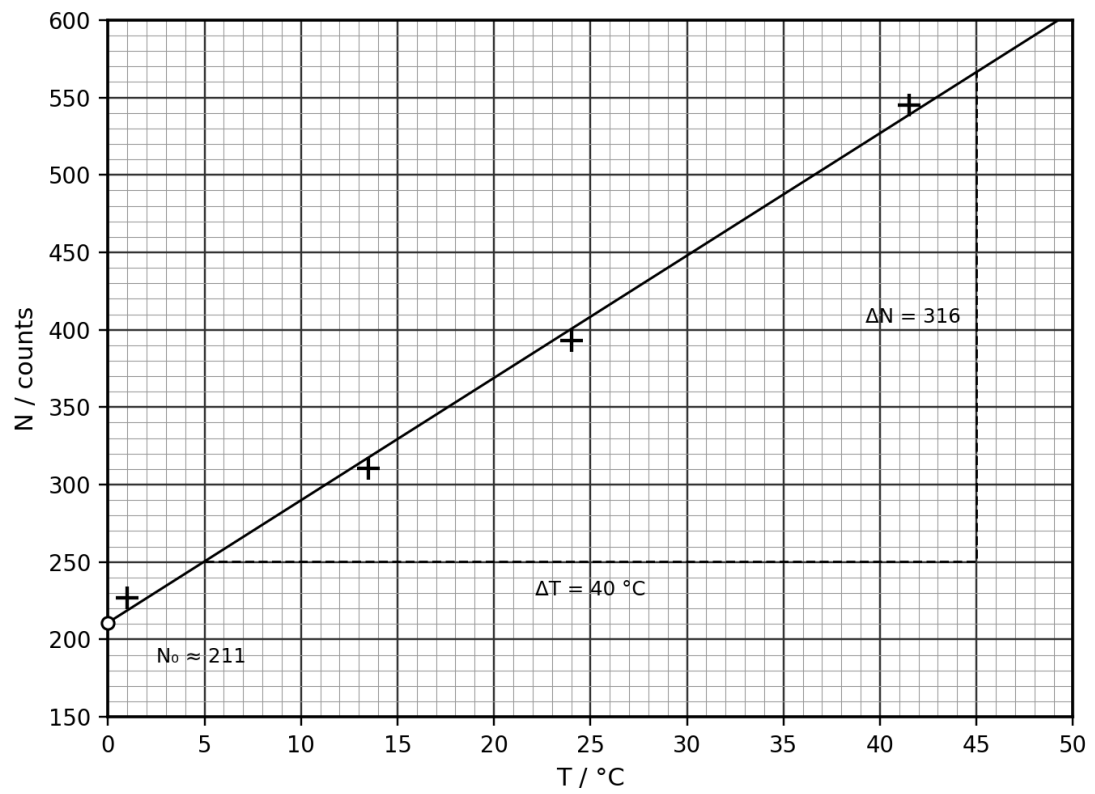
[14]

**Data with the anomalous reading in square brackets and the correct means** (the mean for bath 2 leaves out 271):

| Bath                 | T / °C | N <sub>1</sub> | N <sub>2</sub> | N <sub>3</sub> | N <sub>4</sub> | N <sub>5</sub> | mean N |
|----------------------|--------|----------------|----------------|----------------|----------------|----------------|--------|
| 1 (ice water)        | 1.0    | 230            | 226            | 226            | 226            | 227            | 227.0  |
| 2 (cold water)       | 13.5   | 309            | 309            | 311            | [271]          | 312            | 310.25 |
| 3 (room temperature) | 24.0   | 391            | 394            | 394            | 393            | 393            | 393.0  |
| 4 (hot water)        | 41.5   | 545            | 547            | 547            | 543            | 544            | 545.2  |

**Least-squares line through the four means:**  $m = 7.90 \text{ counts/}^\circ\text{C}$ ,  $N_0 = 210.9$ , so  $T = (N - 210.9) / 7.90$ . A line drawn by hand will give slightly different values ( $m$  within about  $0.5 \text{ counts/}^\circ\text{C}$ ,  $N_0$  within about 8 counts is fine). The test bath ( $N = 483$ ) gives  $T = 34.45 \text{ }^\circ\text{C}$ . The reference thermometer reads  $34.6 \text{ }^\circ\text{C}$ : difference  $0.15 \text{ }^\circ\text{C}$ .

|     | Answer                                                                                                                                                                                                                                                                 | Marks |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (a) | Independent variable: temperature T of the bath (read on the reference thermometer).<br>Dependent variable: thermistor reading N.                                                                                                                                      | 2     |
| (b) | The reading 271 in bath 2 (N <sub>4</sub> ), because it differs from the other four readings of that bath by about 39 counts, while those agree to within 3 counts.                                                                                                    | 1     |
| (c) | Mean of the four remaining readings: 310.25.                                                                                                                                                                                                                           | 1     |
| (d) | Axes labelled "T / °C" (x) and "N / counts" (y). Four points plotted to within half a small square, each marked with a small cross (the centre of the cross is the point).                                                                                             | 3     |
| (e) | A single, straight, thin line drawn with a ruler, passing as close as possible to all the points, with points on both sides of it (not dot-to-dot, not forced through the first and last points).                                                                      | 1     |
| (f) | Triangle drawn with a base of at least half the line (e.g. T = 5 to 45 °C); $m = \Delta N / \Delta T = 7.90 \text{ counts/}^\circ\text{C}$ .                                                                                                                           | 2     |
| (g) | $N_0 \approx 211$ (read at T = 0 °C; fit: 210.9). Equation $T = (N - 210.9) / 7.90$ .                                                                                                                                                                                  | 1     |
| (h) | $T = (483 - 210.9) / 7.90 \approx 34.5 \text{ }^\circ\text{C}$ (accepted: 33.0–36.0 °C).                                                                                                                                                                               | 1     |
| (i) | Meets the requirement: $ 34.6 - 34.5  = 0.15 \text{ }^\circ\text{C} < 1 \text{ }^\circ\text{C}$ .                                                                                                                                                                      | 1     |
| (j) | One specific improvement, e.g.: take several readings in the test bath; use a reference thermometer with smaller divisions (0.1 °C resolution); stir the bath and wait for equilibrium before reading; use more calibration baths; use a second independent test bath. | 1     |



Reference graph: points = means, least-squares line, gradient triangle from  $T = 5$  to  $T = 45 ^{\circ}\text{C}$ , intercept  $N_0$  on the axis.

### III · Structured questions

[14]

|      | Answer                                                                                                                                                                                                                                                                                                                                 | Marks |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(a) | (i) $37\text{ }^{\circ}\text{C} = 310\text{ K}$ ; (ii) $90\text{ K} = -183\text{ }^{\circ}\text{C}$ .                                                                                                                                                                                                                                  | 2     |
| 1(b) | Heat is not something a body has. Temperature is a property (state quantity) of the body; heat is energy transferred because of a temperature difference. What the body has is internal energy. [1] heat = energy in transfer because of $\Delta T$ (a process, not stored). [1] the body has a temperature / internal energy instead. | 2     |
| 1(c) | 0 K (absolute zero) is the lowest possible temperature; nothing can be cooled below it, so $-10\text{ K}$ does not exist.                                                                                                                                                                                                              | 1     |
| 2(a) | $0\text{ }^{\circ}\text{C}$ : melting point of pure ice. $100\text{ }^{\circ}\text{C}$ : boiling point of pure water (at standard atmospheric pressure).                                                                                                                                                                               | 2     |
| 2(b) | The glass bulb warms up first and expands, so the space inside the bulb increases and the liquid level falls (1). Then the liquid warms up and expands more than the glass, so the column rises (1).                                                                                                                                   | 2     |
| 3(a) | $\Delta T = 65 - (15) = 50\text{ }^{\circ}\text{C}$ [1]; $\Delta L = \alpha L_0 \Delta T = 23.0 \times 10^{-6} \times 2.5 \times 50$ [1] = $2.875 \times 10^{-3}\text{ m} = 2.9\text{ mm}$ [1].                                                                                                                                        | 3     |
| 3(b) | $\beta = 2\alpha = 46.0 \times 10^{-6}\text{ }^{\circ}\text{C}^{-1}$ [1]; $\Delta A = \beta A_0 \Delta T = 46.0 \times 10^{-6} \times 0.80 \times 50 = 1.84 \times 10^{-3}\text{ m}^2$ (18.4 $\text{cm}^2$ ) [1].                                                                                                                      | 2     |

#### IV · Problema — Calettamento a caldo di un anello

[8]

|     | Answer                                                                                                                                                                                                                                                                                                                                   | Marks |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (a) | Aumenta: il foro si dilata come se fosse fatto dello stesso materiale (come un disco pieno); tutte le lunghezze del pezzo scalano con la stessa frazione.                                                                                                                                                                                | 1     |
| (b) | Diametro minimo del foro: $50,06 + 0,04 = 50,10$ mm [1]. $\Delta D = 0,10$ mm, quindi $\Delta T = \Delta D / (\alpha D) = 0,10 / (12,0 \times 10^{-6} \times 50,00) = 166,7$ °C [1]. $T = 20 + 166,7 = 186,7$ °C, minore del limite di 450 °C: compatibile [1].                                                                          | 3     |
| (c) | Ogni lunghezza cresce di $\Delta D/D = 0,200$ %; l'area, che dipende dal quadrato del diametro, cresce circa del doppio: $\Delta A/A \approx 2 \alpha \Delta T = \beta \Delta T = 0,40$ % [1 per $\beta = 2\alpha$ , 1 per il risultato]. Non serve conoscere il diametro ( $\Delta A \approx 7,85$ mm <sup>2</sup> per $D = 50,00$ mm). | 2     |
| (d) | Interferenza a 20 °C: $50,06 - 50,00 = 0,06$ mm. A 120 °C: albero 50,1551 mm, foro 50,0600 mm, interferenza 0,0951 mm [1]. Il collegamento si stringe (+0,0351 mm) perché l'ottone (albero) ha $\alpha$ maggiore dell'acciaio (anello): l'albero cresce più del foro [1].                                                                | 2     |

**Dati utili e dati di disturbo.** Servono: diametri, gioco richiesto, coefficienti  $\alpha$ , limite del costruttore (per la verifica in (b)), temperatura di esercizio (per (d)). Non servono: massa, altezza o spessore del pezzo, potenza del forno, tempo di riscaldamento, pezzi prodotti, giri/min.